Genome-wide fusion-hotspot analysis of msk_impact_50k_2026

Abstract

This manuscript synthesizes a genome-wide fusion-hotspot cohort scan of msk_impact_50k_2026: 3919 genes carried at least one structural-variant record in the cohort, of which 544 passed the >= 5-distinct-patient recurrence gate and were analyzed with the full registered algorithm suite (4 using hand-curated gene configs, 520 auto-configured, 20 gated in but unresolvable). 1 gene reached genome-wide Benjamini-Hochberg FDR significance (q < 0.05): ETV6 (90 events, 71.1% in-frame, 75.6% domain-retained, Fisher p=5.12326e-06, q=0.00433427). 15 additional genes form a highly ranked non-FDR-significant tier flagged for targeted follow-up (see Honorable mentions, below).

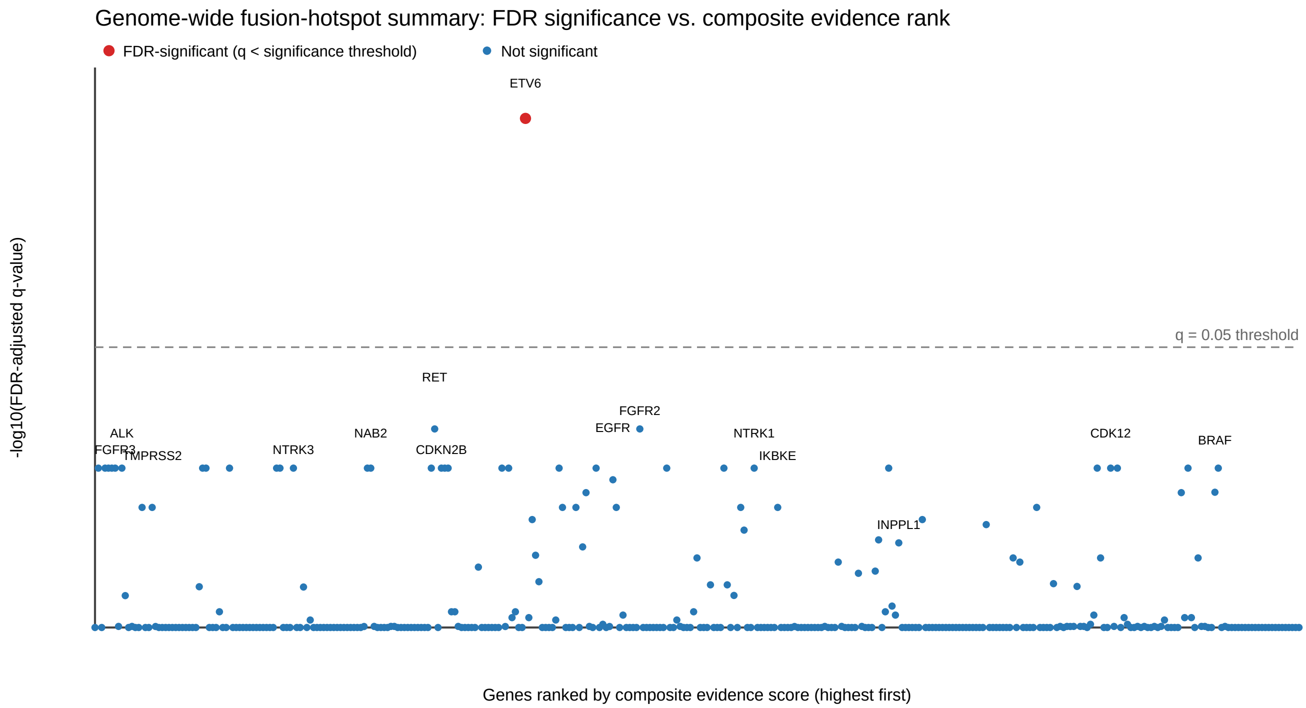
Methods

Structural-variant records were retrieved from the msk_impact_50k_2026 cBioPortal study and gated to genes with at least 5 distinct patient(s) carrying a structural-variant record (544 of 3919 genes passed the gate: 4 using hand-curated gene configs and 520 auto-configured from Genome Nexus canonical-transcript/Pfam-domain data, with 20 gated-in gene(s) left unresolvable). Each gated gene was analyzed with the full registered algorithm suite (composite_score, confidence_stats, cutpoint_detection, domain_disruption, domain_retention, exon_retention, frequency, joint_partner). Domain-retention and domain-disruption significance were assessed per gene with Fisher's exact test and a breakpoint-position permutation test; the resulting p-values across all 544 scanned genes were jointly corrected with Benjamini-Hochberg false-discovery-rate correction at q < 0.05.

Results

Genome-wide summary

Genome-wide summary of 359 scanned genes with an FDR-adjusted q-value, ranked left-to-right by composite evidence score; the dashed line marks the q=0.05 significance threshold, with 1 gene above it.


Genome-wide fusion-hotspot summary: FDR significance vs. composite evidence rank
FDR-significant (q < significance threshold)
Not significant
ETV6
q = 0.05 threshold
RET
FGFR2
EGFR
ALK
NAB2
NTRK1
CDK12
BRAF
FGFR3
TMPRSS2
NTRK3
CDKN2B
IKBKE
INPPL1
-log10(FDR-adjusted q-value)
Genes ranked by composite evidence score (highest first)

FDR-significant and honorable-mention genes

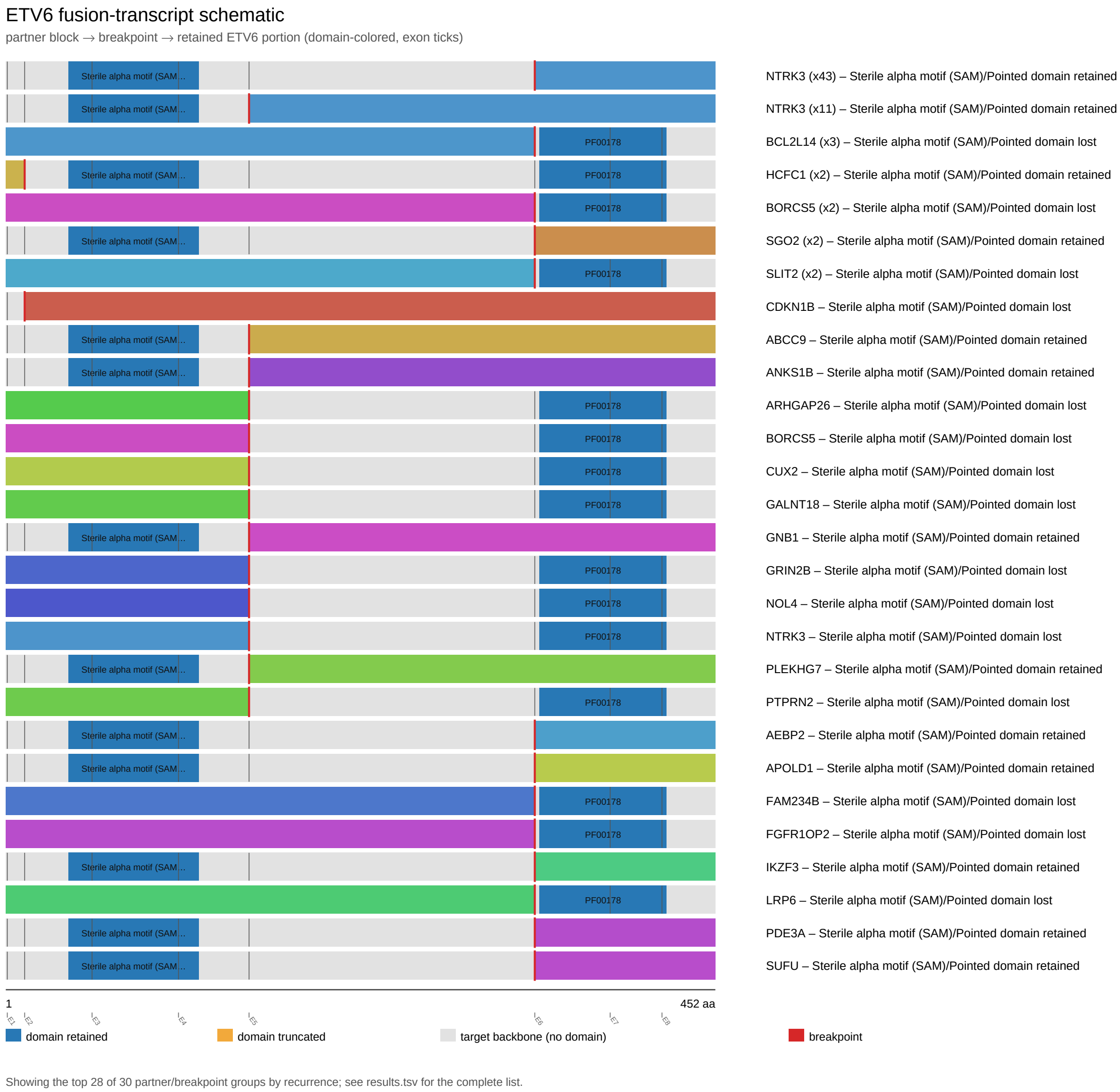
Gene	Tier	N events	In-frame%	Domain-ret.%	Fisher p	Best q
ETV6	FDR-significant	90	71.11	75.56	5.123e-06	0.004334
RET	Honorable mention	194	75.26	92.27	0.0004197	0.1198
FGFR2	Honorable mention	136	79.41	84.56	0.0004247	0.1198
ALK	Honorable mention	272	83.09	95.96	0.001917	0.1821
EGFR	Honorable mention	55	50.91	74.55	0.01145	0.2062
BRAF	Honorable mention	179	84.36	91.06	0.01337	0.2356
FGFR3	Honorable mention	152	48.68	93.42	0.01948	0.1821
CDKN2B	Honorable mention	12	16.67	16.67	0.02222	0.1821
IKBKE	Honorable mention	12	25	33.33	0.02424	0.2771
TMPRSS2	Honorable mention	867	29.99	6.113	0.0292	0.2771
NTRK1	Honorable mention	78	41.03	82.05	0.04826	0.1821
INPPL1	Honorable mention	14	28.57	50	0.04895	0.4047
CDK12	Honorable mention	43	16.28	37.21	0.0677	0.1821
NTRK3	Honorable mention	58	94.83	94.83	0.06955	0.1821
NAB2	Honorable mention	72	34.72	68.06	0.1126	0.1821
KDM5A	Honorable mention	10	30	50	0.119	0.8758

Gene highlights

ETV6 (FDR-significant)

ETV6 was analyzed across 90 fusion events, 71.1% in-frame and 75.6% domain-retained. Domain-retention Fisher's exact test p=5.12326e-06, genome-wide BH-adjusted q=0.00433427 (statistically significant at alpha=0.05).

Reused from the ETV6 individual gene report (fusion_schematic.svg).

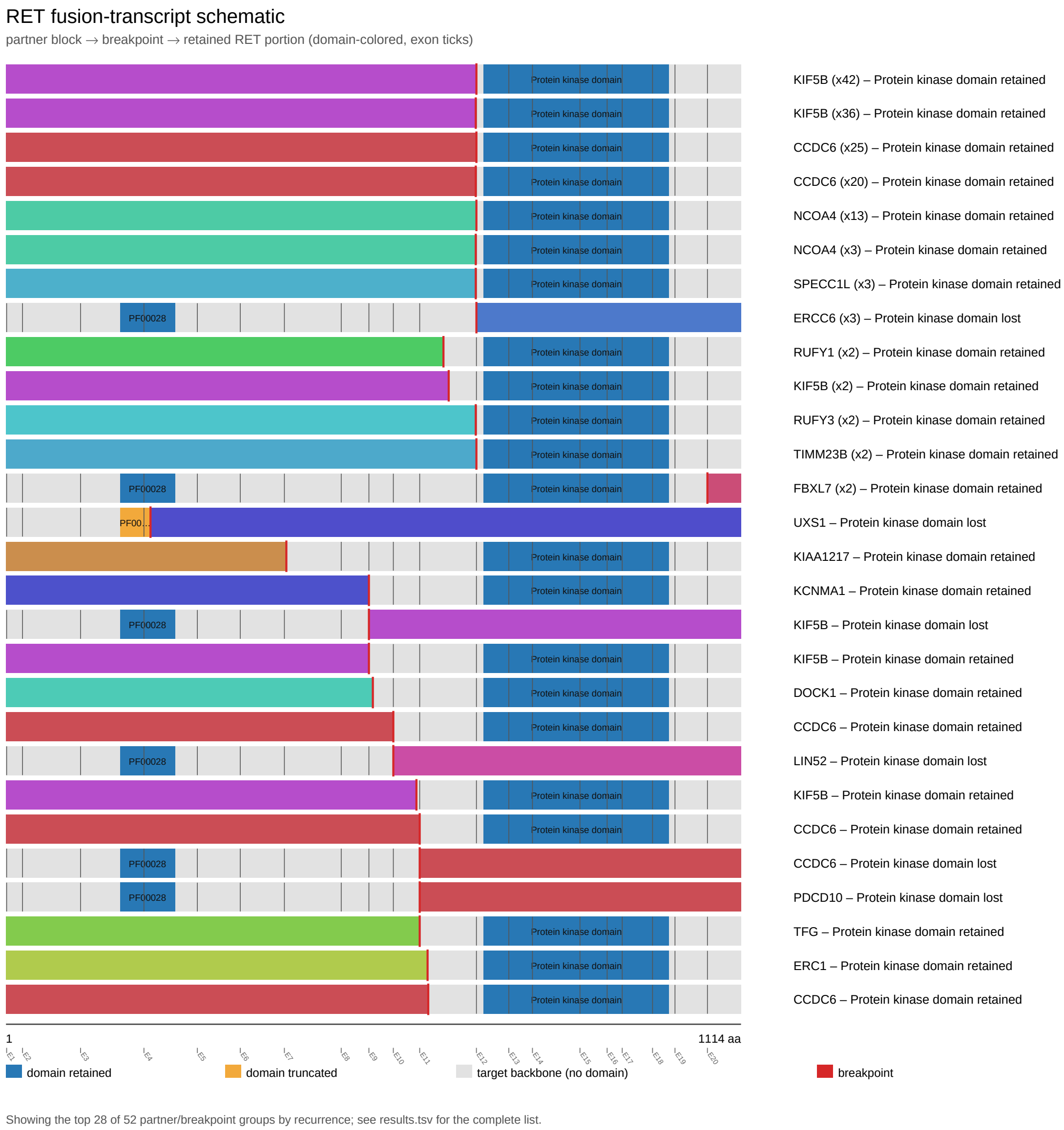


Full per-gene detail: gene_reports/etv6/report.md

RET (Honorable mention, Curated gene config)

RET was analyzed across 194 fusion events, 75.3% in-frame and 92.3% domain-retained. Domain-retention Fisher's exact test p=0.000419666, genome-wide BH-adjusted q=0.119762 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the RET individual gene report (fusion_schematic.svg).

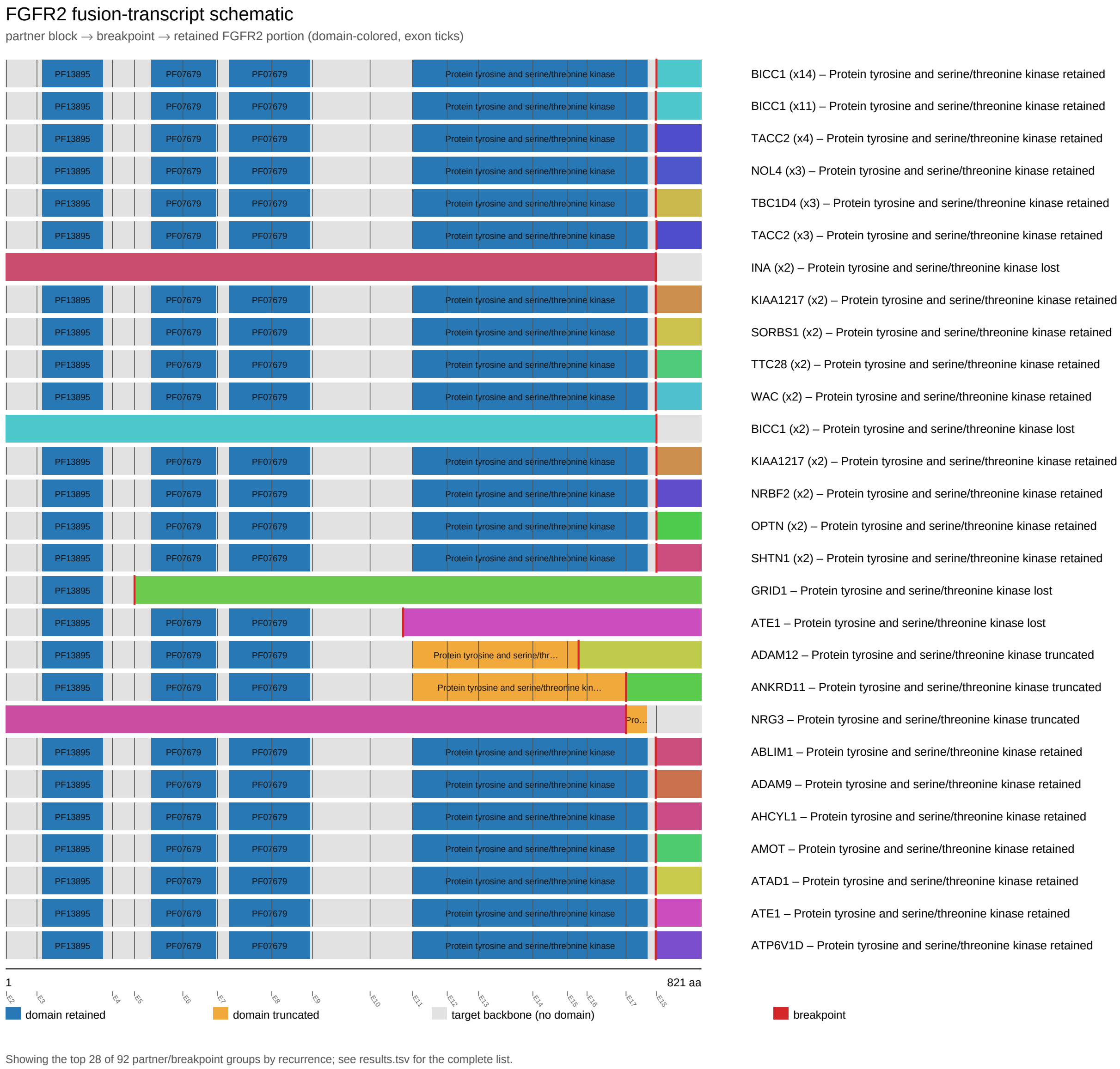


Full per-gene detail: gene_reports/ret/report.md

FGFR2 (Honorable mention)

FGFR2 was analyzed across 136 fusion events, 79.4% in-frame and 84.6% domain-retained. Domain-retention Fisher's exact test p=0.000424687, genome-wide BH-adjusted q=0.119762 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FGFR2 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/fgfr2/report.md

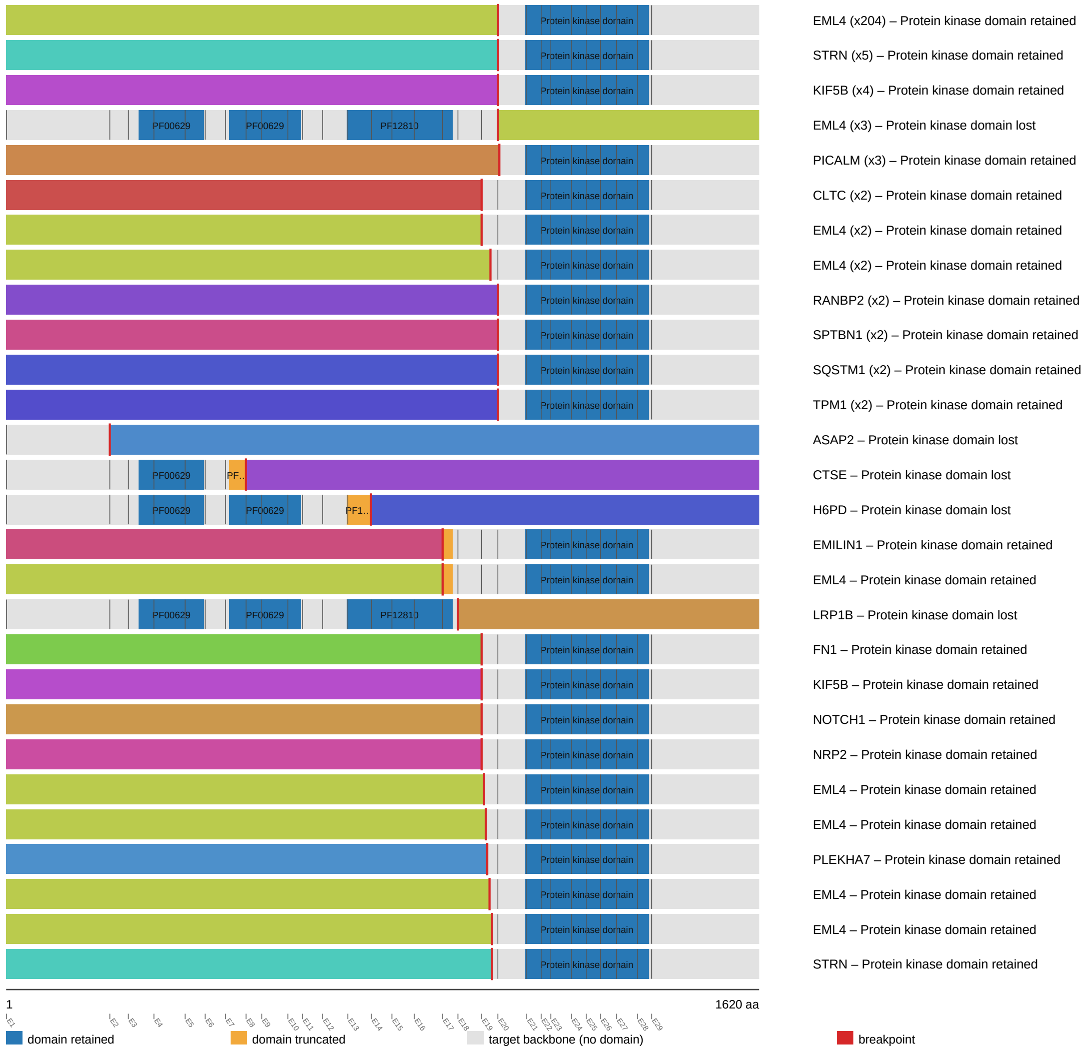
ALK (Honorable mention, Curated gene config)

ALK was analyzed across 272 fusion events, 83.1% in-frame and 96.0% domain-retained. Domain-retention Fisher's exact test p=0.00191661, genome-wide BH-adjusted q=0.182092 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the ALK individual gene report (fusion_schematic.svg).

ALK fusion-transcript schematic

partner block → breakpoint → retained ALK portion (domain-colored, exon ticks)



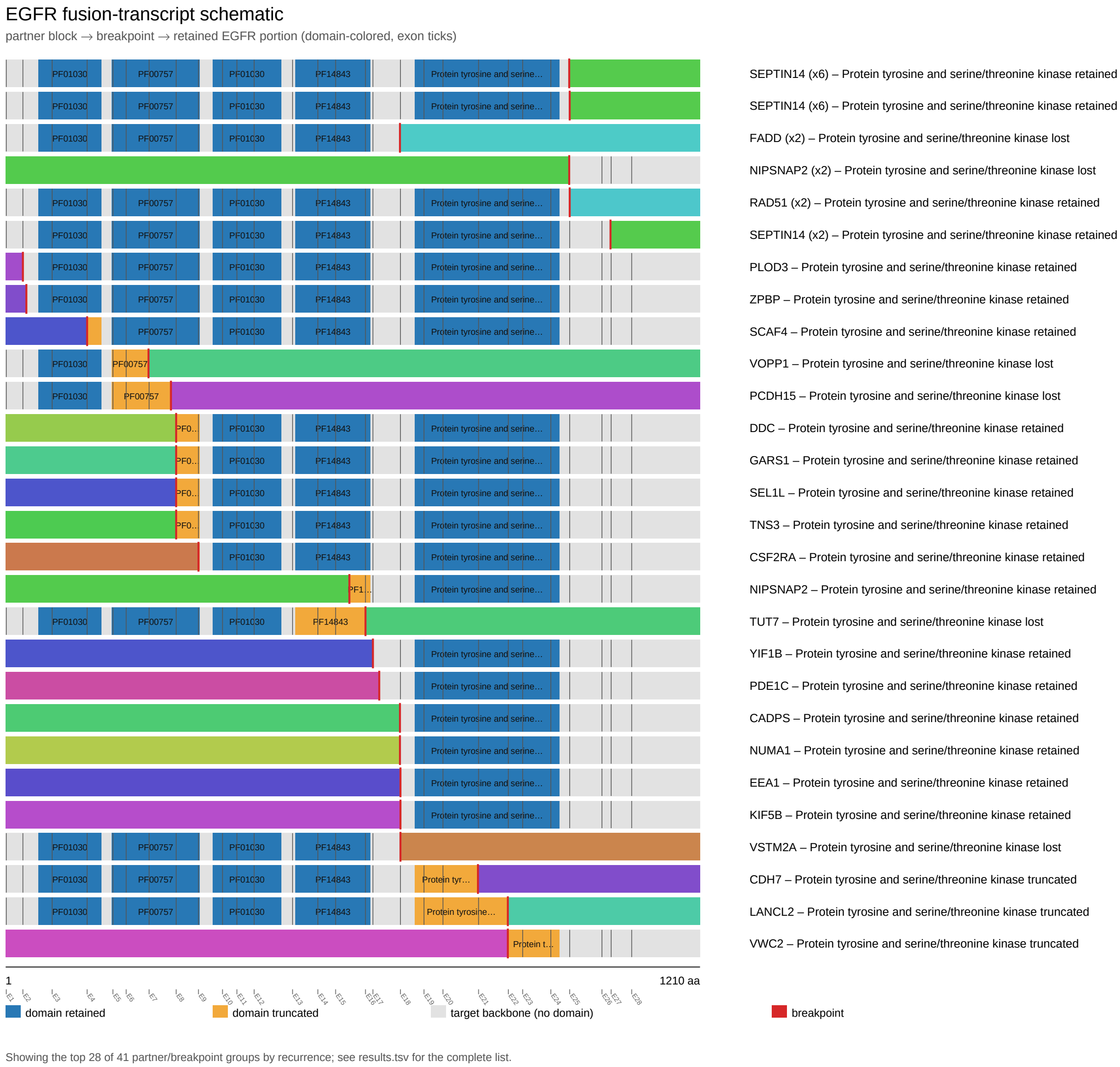
Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/alk/report.md

EGFR (Honorable mention)

EGFR was analyzed across 55 fusion events, 50.9% in-frame and 74.5% domain-retained. Domain-retention Fisher's exact test p=0.0114539, genome-wide BH-adjusted q=0.206169 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the EGFR individual gene report (fusion_schematic.svg).

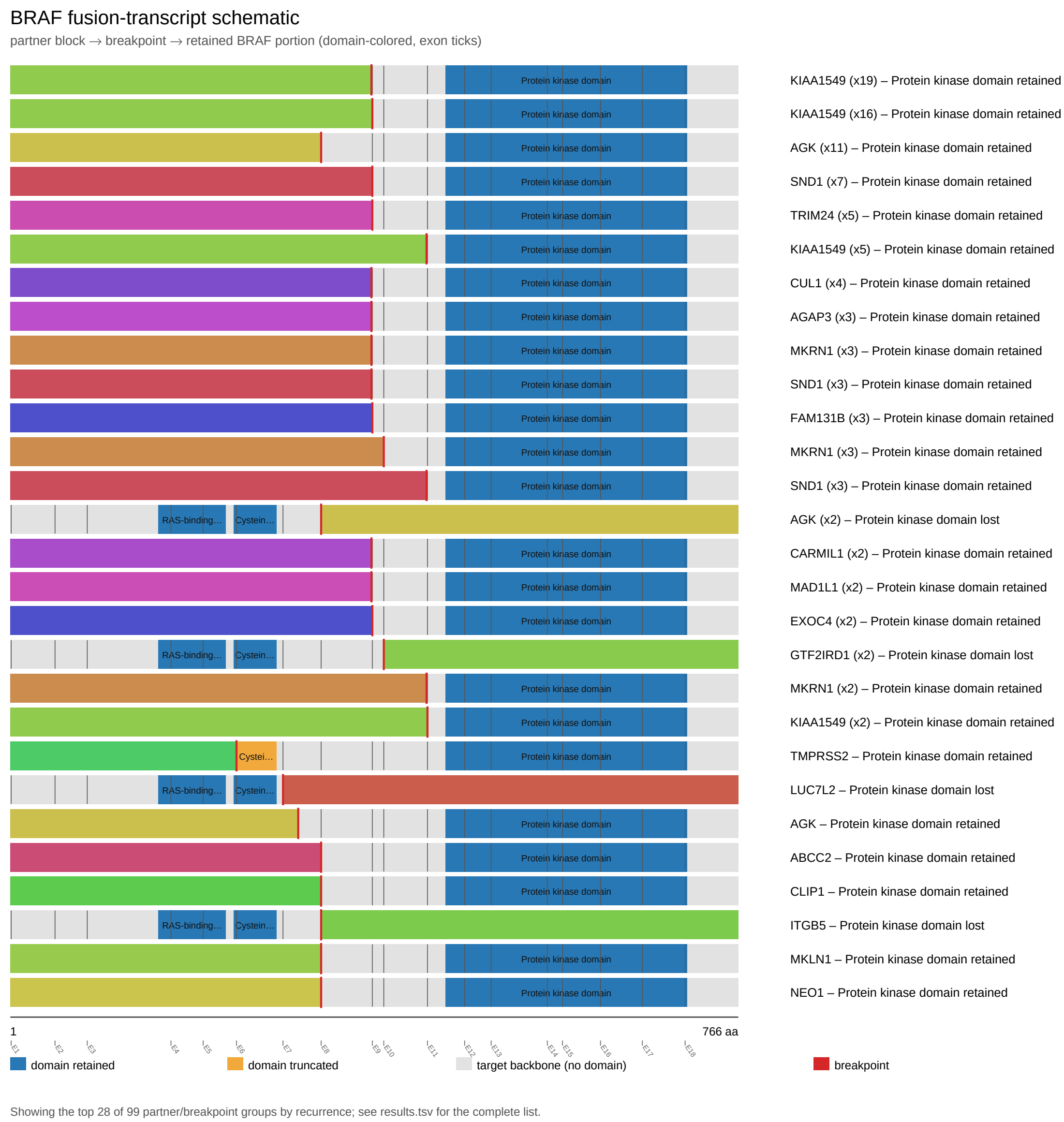


Full per-gene detail: gene_reports/egfr/report.md

BRAF (Honorable mention, Curated gene config)

BRAF was analyzed across 179 fusion events, 84.4% in-frame and 91.1% domain-retained. Domain-retention Fisher's exact test p=0.0133676, genome-wide BH-adjusted q=0.235603 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the BRAF individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/braf/report.md

FGFR3 (Honorable mention)

FGFR3 was analyzed across 152 fusion events, 48.7% in-frame and 93.4% domain-retained. Domain-retention Fisher's exact test p=0.0194824, genome-wide BH-adjusted q=0.182092 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the FGFR3 individual gene report (fusion_schematic.svg).

FGFR3 fusion-transcript schematic

partner block → breakpoint → retained FGFR3 portion (domain-colored, exon ticks)

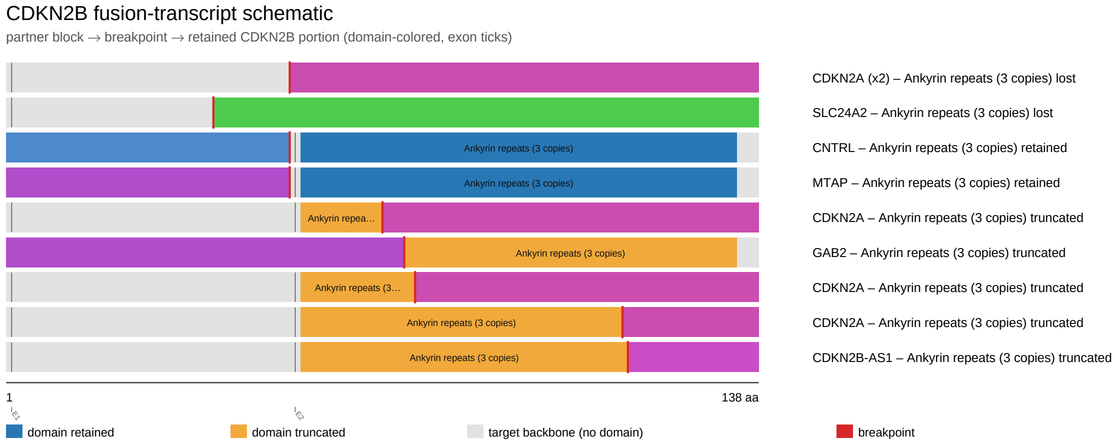
Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/fgfr3/report.md

CDKN2B (Honorable mention)

CDKN2B was analyzed across 12 fusion events, 16.7% in-frame and 16.7% domain-retained. Domain-retention Fisher's exact test p=0.0222222, genome-wide BH-adjusted q=0.182092 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDKN2B individual gene report (fusion_schematic.svg).

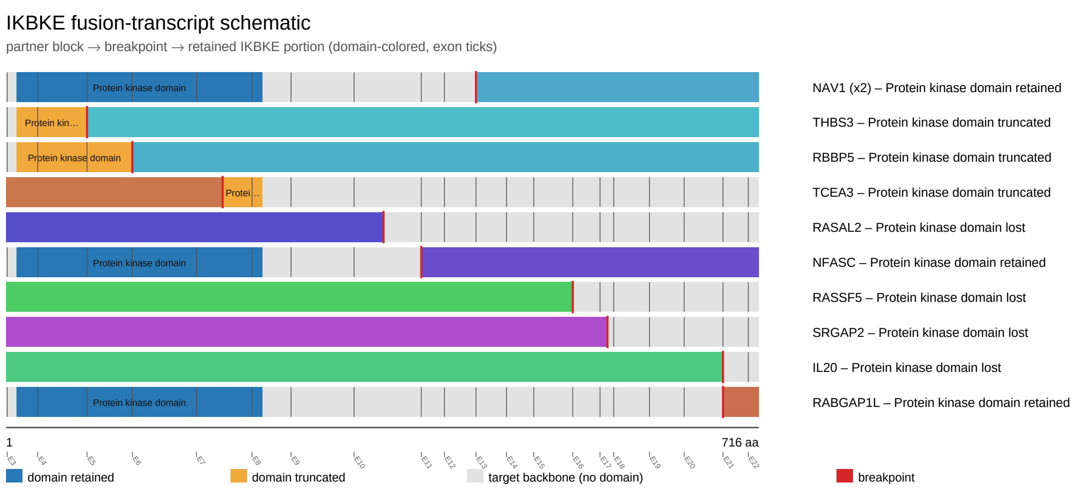


Full per-gene detail: gene_reports/cdkn2b/report.md

IKBKE (Honorable mention)

IKBKE was analyzed across 12 fusion events, 25.0% in-frame and 33.3% domain-retained. Domain-retention Fisher's exact test p=0.0242424, genome-wide BH-adjusted q=0.2771 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the IKBKE individual gene report (fusion_schematic.svg).

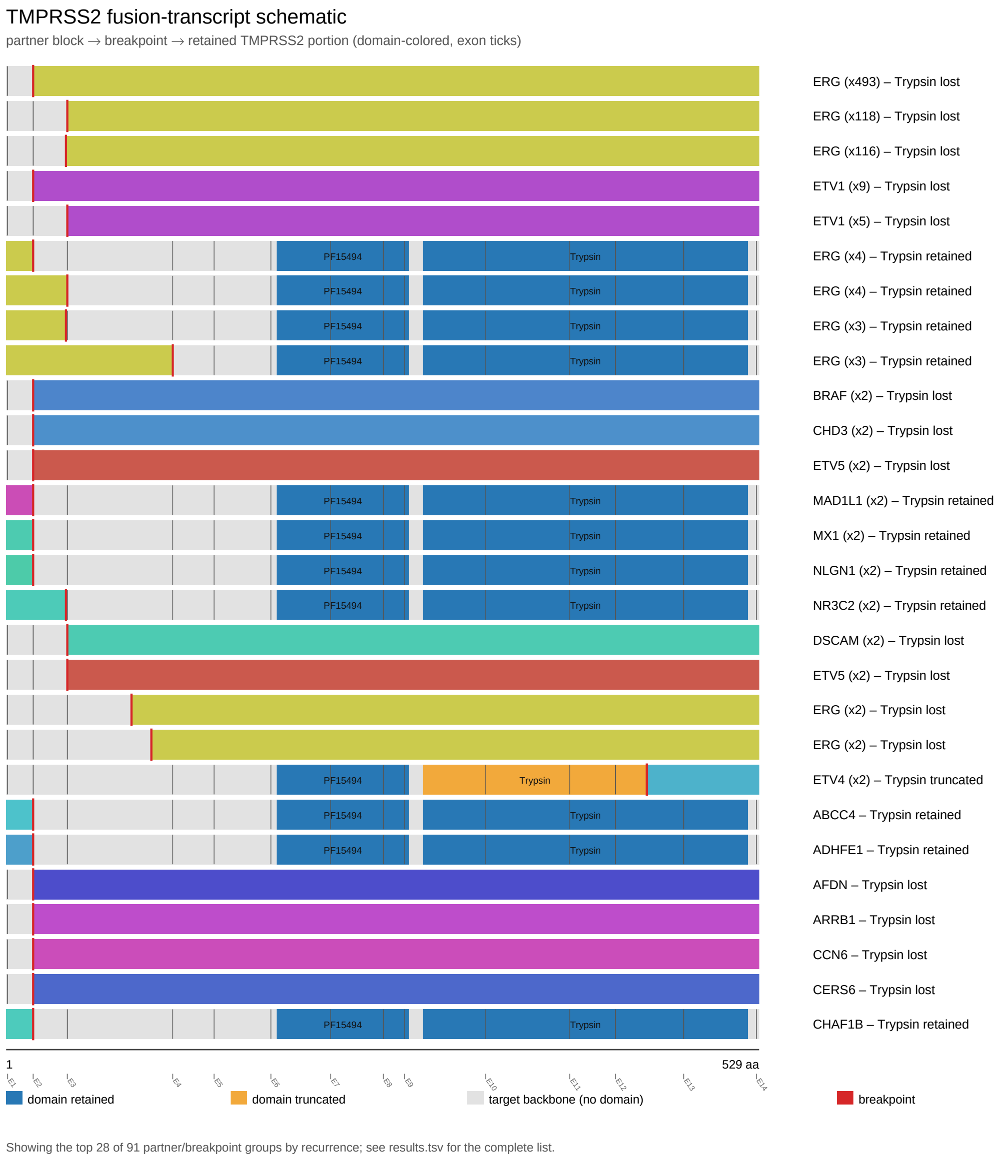


Full per-gene detail: gene_reports/ikbke/report.md

TMPRSS2 (Honorable mention)

TMPRSS2 was analyzed across 867 fusion events, 30.0% in-frame and 6.1% domain-retained. Domain-retention Fisher's exact test p=0.0292045, genome-wide BH-adjusted q=0.2771 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the TMPRSS2 individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/tmprss2/report.md

NTRK1 (Honorable mention, Curated gene config)

NTRK1 was analyzed across 78 fusion events, 41.0% in-frame and 82.1% domain-retained. Domain-retention Fisher's exact test p=0.0482628, genome-wide BH-adjusted q=0.182092 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

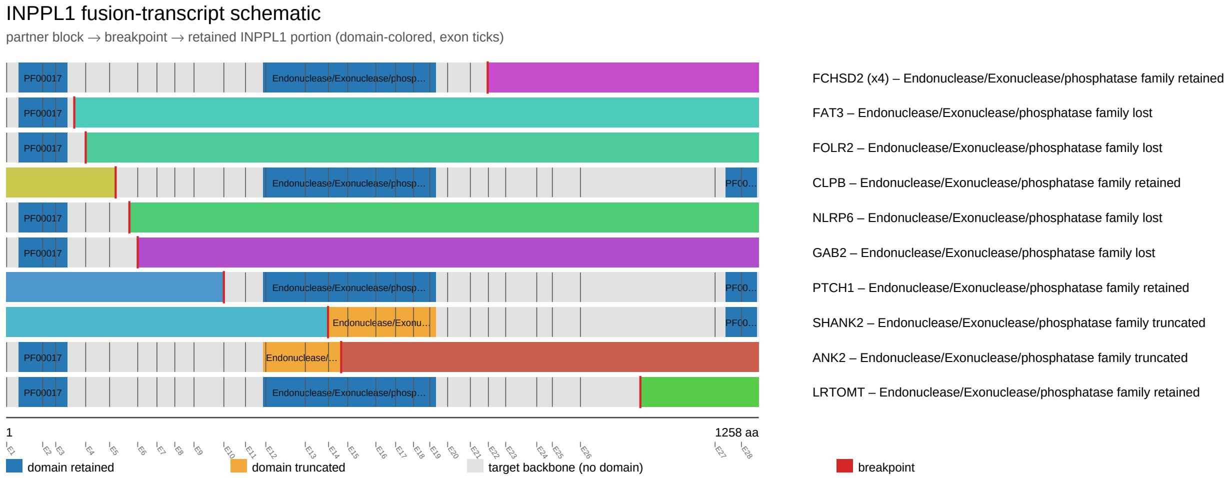
Reused from the NTRK1 individual gene report (fusion_schematic.svg).

Full per-gene detail: gene_reports/ntrk1/report.md

INPPL1 (Honorable mention)

INPPL1 was analyzed across 14 fusion events, 28.6% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test p=0.048951, genome-wide BH-adjusted q=0.404722 (statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the INPPL1 individual gene report (fusion_schematic.svg).

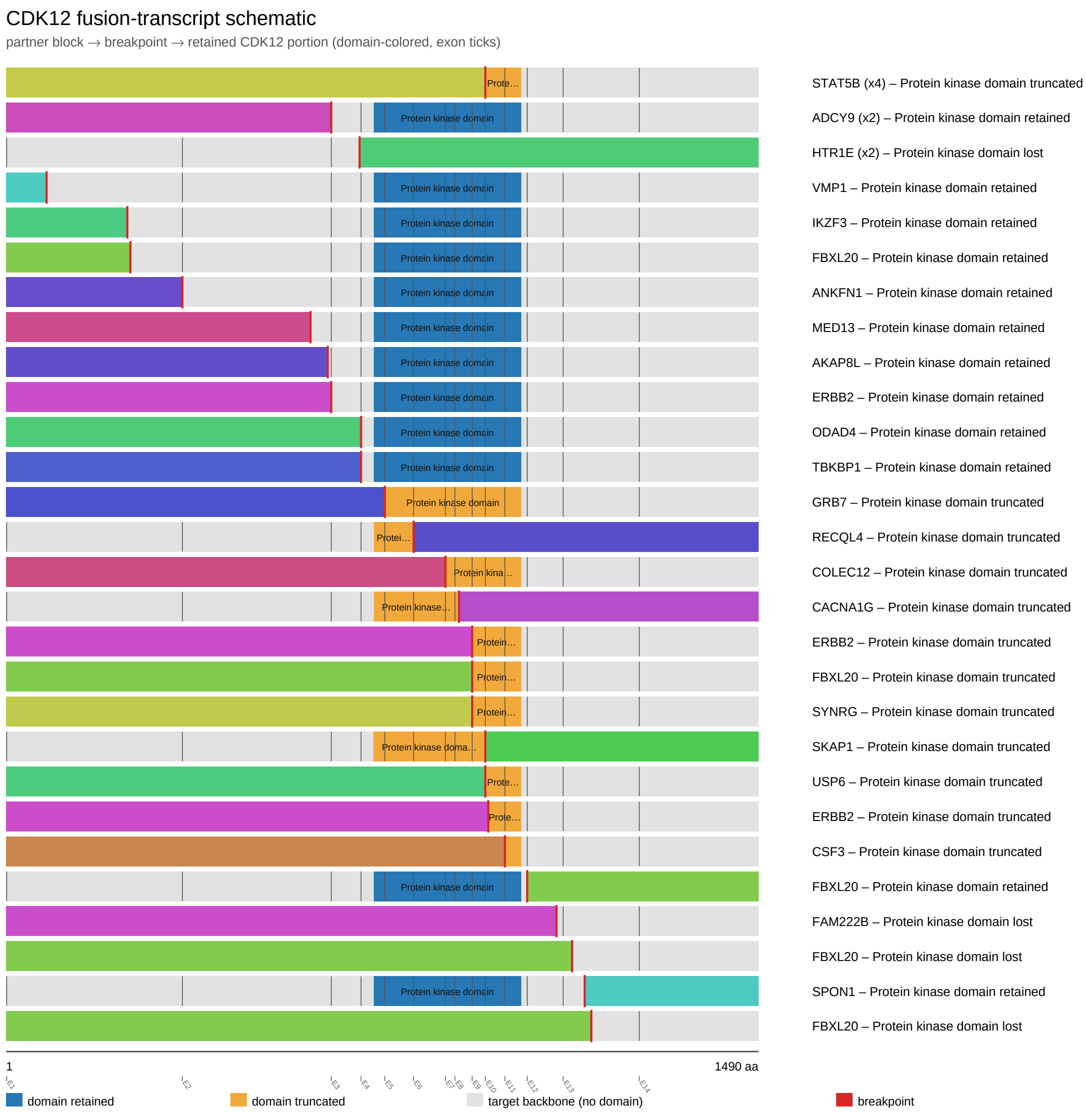


Full per-gene detail: gene_reports/inppl1/report.md

CDK12 (Honorable mention)

CDK12 was analyzed across 43 fusion events, 16.3% in-frame and 37.2% domain-retained. Domain-retention Fisher's exact test p=0.0677006, genome-wide BH-adjusted q=0.182092 (not statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the CDK12 individual gene report (fusion_schematic.svg).



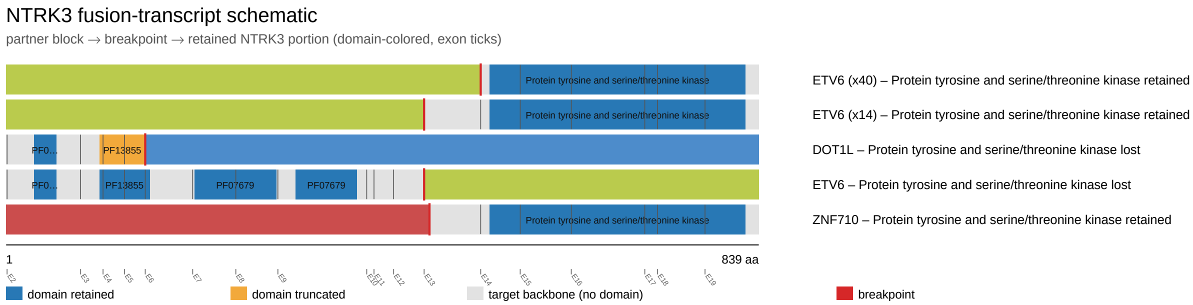
Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Full per-gene detail: gene_reports/cdk12/report.md

NTRK3 (Honorable mention)

NTRK3 was analyzed across 58 fusion events, 94.8% in-frame and 94.8% domain-retained. Domain-retention Fisher's exact test p=0.0695489, genome-wide BH-adjusted q=0.182092 (not statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NTRK3 individual gene report (fusion_schematic.svg).

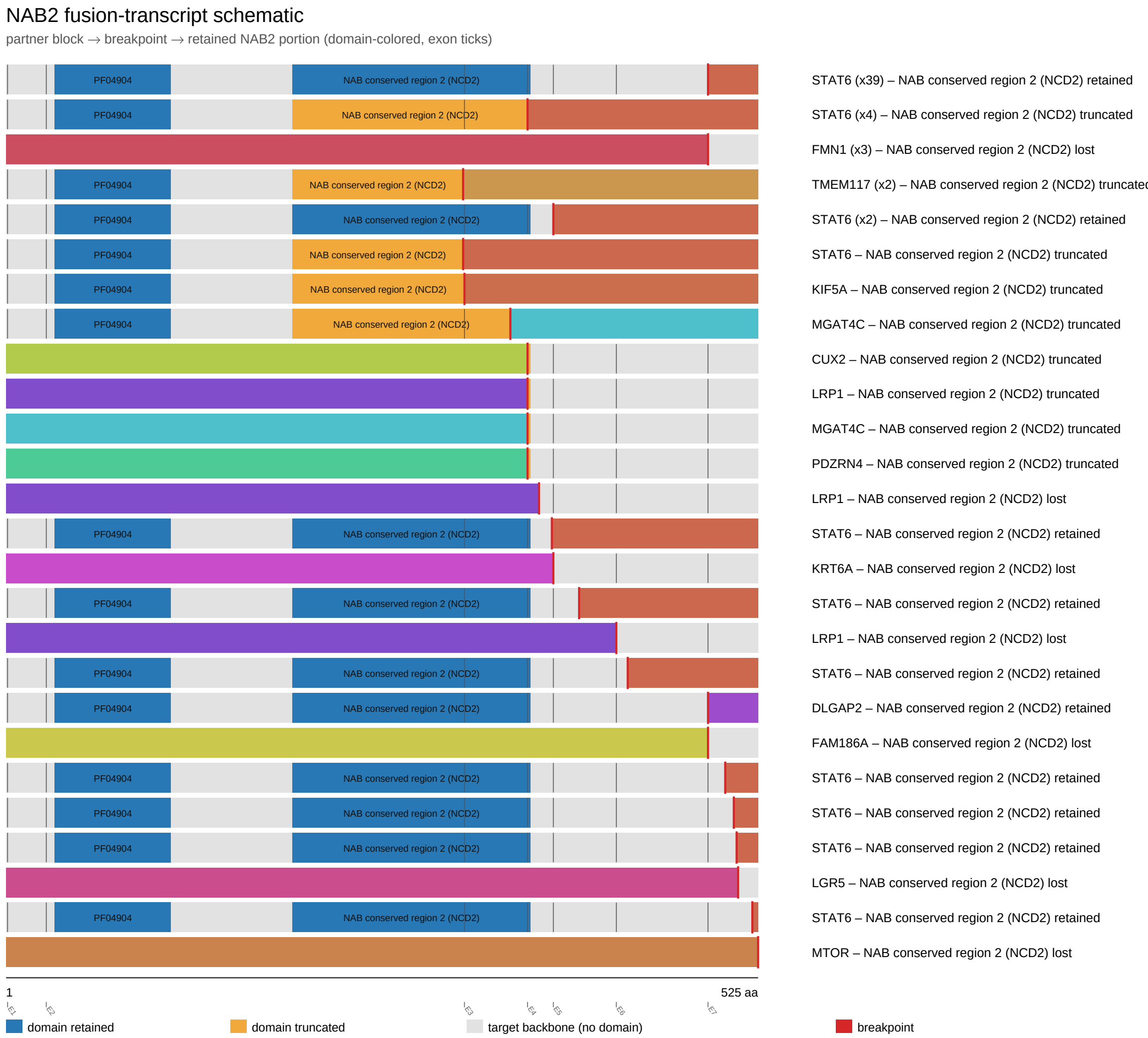


Full per-gene detail: gene_reports/ntrk3/report.md

NAB2 (Honorable mention)

NAB2 was analyzed across 72 fusion events, 34.7% in-frame and 68.1% domain-retained. Domain-retention Fisher's exact test p=0.11258, genome-wide BH-adjusted q=0.182092 (not statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the NAB2 individual gene report (fusion_schematic.svg).

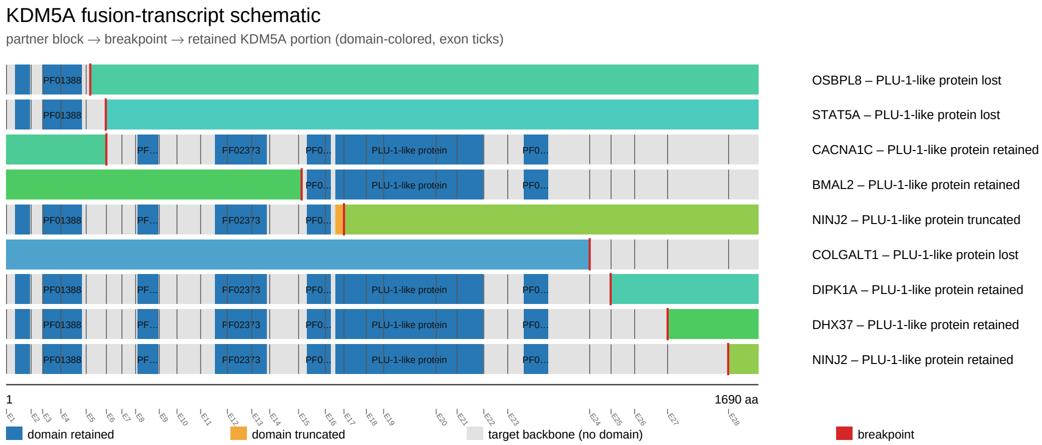


Full per-gene detail: gene_reports/nab2/report.md

KDM5A (Honorable mention)

KDM5A was analyzed across 10 fusion events, 30.0% in-frame and 50.0% domain-retained. Domain-retention Fisher's exact test p=0.119048, genome-wide BH-adjusted q=0.875776 (not statistically significant at alpha=0.05). Did not survive genome-wide multiple-testing correction (FDR-adjusted q-value at or above the significance threshold), but ranks highly by raw p-value among the non-FDR-significant genes and may warrant targeted follow-up. This is NOT a claim of statistical significance.

Reused from the KDM5A individual gene report (fusion_schematic.svg).



Full per-gene detail: gene_reports/kdm5a/report.md

Discussion

• Cross-gene Benjamini-Hochberg FDR correction was applied jointly across all 544 scanned genes' p-values. This reduces false-positive findings relative to testing each gene in isolation, but is a conservative correction: a real per-gene effect can fail to reach the q < 0.05 threshold once corrected across the full scanned gene set.

• 4 gene(s) used hand-curated gene configurations, while 520 gene(s) were auto-configured from Genome Nexus canonical-transcript/Pfam-domain data using a kinase/catalytic-keyword heuristic to select the tracked domain; auto-configured domains have not been manually verified the way hand-curated ones have.

• Data were retrieved live from the public msk_impact_50k_2026 cBioPortal study as of 2026-09-04T15:54:53.136605+00:00. Some cBioPortal cohorts (e.g. actively accruing clinical-sequencing panels such as MSK-IMPACT) are updated periodically, so exact counts could shift if this scan is re-run later against such a cohort.

• All findings in this manuscript are computational, hypothesis-generating candidate evidence from bioinformatic analysis of public cohort data, not validated clinical calls; a gene's presence here is not a therapeutic or diagnostic recommendation.

Appendix: per-gene report index

Gene	Tier	Config source	N events	Best q	Full report
ETV6	FDR-significant	auto	90	0.004334	gene_reports/etv6/report.md
RET	Honorable mention; Curated gene config	curated	194	0.1198	gene_reports/ret/report.md
FGFR2	Honorable mention	auto	136	0.1198	gene_reports/fgfr2/report.md
ALK	Honorable mention; Curated gene config	curated	272	0.1821	gene_reports/alk/report.md
EGFR	Honorable mention	auto	55	0.2062	gene_reports/egfr/report.md
BRAF	Honorable mention; Curated gene config	curated	179	0.2356	gene_reports/braf/report.md
FGFR3	Honorable mention	auto	152	0.1821	gene_reports/fgfr3/report.md
CDKN2B	Honorable mention	auto	12	0.1821	gene_reports/cdkn2b/report.md
IKBKE	Honorable mention	auto	12	0.2771	gene_reports/ikbke/report.md
TMPRSS2	Honorable mention	auto	867	0.2771	gene_reports/tmprss2/report.md
NTRK1	Honorable mention; Curated gene config	curated	78	0.1821	gene_reports/ntrk1/report.md
INPPL1	Honorable mention	auto	14	0.4047	gene_reports/inppl1/report.md
CDK12	Honorable mention	auto	43	0.1821	gene_reports/cdk12/report.md
NTRK3	Honorable mention	auto	58	0.1821	gene_reports/ntrk3/report.md
NAB2	Honorable mention	auto	72	0.1821	gene_reports/nab2/report.md
KDM5A	Honorable mention	auto	10	0.8758	gene_reports/kdm5a/report.md